

LESSON 5-8 • HOMEWORK PACKET • SHEET 1 OF 2**Surface Area of Pyramids**

Maryland Standard 6.G.A.4 • Learning target – I can find the surface area of a pyramid by adding the base area and the areas of its triangular faces.

REMEMBER

- SA of a pyramid = **base area** + area of all the **triangular faces**.
- Each triangle's height is the **slant height**, not the height of the pyramid.

WORKED EXAMPLE

Square pyramid: base 8 cm, slant height 7 cm.

Step 1 Base: $8 \times 8 = 64$

Step 2 One triangle: $\frac{1}{2} \times 8 \times 7 = 28$; four: 112

Step 3 Add: $64 + 112 = 176$

Answer 176 cm²

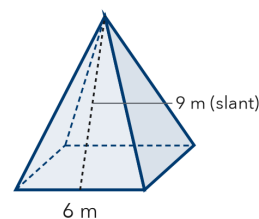
PRACTICE

show your work • label units

1. Find the surface area of the square pyramid.

Show your work.

Answer _____ m²

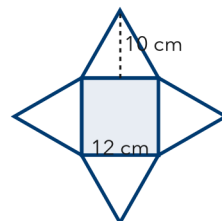


2. Use the net of the square pyramid.

a. Area of the base _____ cm²

b. Area of one triangle _____ cm²

Surface area _____ cm²



3. Find the surface area of each square pyramid.

a. base 4 in, slant height 5 in _____ in²

b. base 10 cm, slant height 12 cm _____ cm²

4. **Gift Box.** A gift box is a square pyramid with a base 9 in on each side and a slant height of 11 in. How much paper covers every face?

Show your work.

Answer _____ in²

5. **Find the Error.** A square pyramid has a base 6 ft on each side, a height of 4 ft, and a slant height of 5 ft. Kofi wrote $SA = 36 + 4 \times (\frac{1}{2} \times 6 \times 4) = 84$ ft². Explain his mistake and find the surface area.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about # _____

AT HOME Use a folded paper tent to show someone a pyramid's height and its slant height.

LESSON 5-8 • HOMEWORK PACKET • SHEET 2 OF 2**Lateral Area of Pyramids**

Maryland Standard 6.G.A.4 • Learning target – I can find the lateral area of a pyramid, which covers only its slanted faces.

REMEMBER

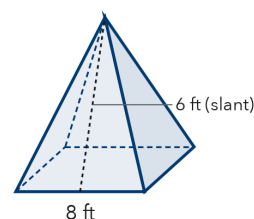
- **Lateral area** is the area of the slanted faces only. Do **not** include the base.
- Lateral area = surface area – base area.

WORKED EXAMPLE

Square pyramid: base 10 m, slant height 13 m. Find the lateral area.

Step 1 One face: $\frac{1}{2} \times 10 \times 13 = 65$ **Step 2** Four faces: $4 \times 65 = 260$ **Answer** **260 m²****PRACTICE***show your work • label units*

1. Find the lateral area of the square pyramid.

*Show your work.***Answer** _____ ft²

2. Find each lateral area.

a. SA = 200 in², base = 64 in² _____ in²**b.** 4 faces, each 18 cm² _____ cm²**c.** square base, side 6 m; slant height 10 m _____ m²**d.** square base, side 3 ft; slant height 4 ft _____ ft²

- 3.
- Tower Roof.**
- A roof is a square pyramid with a base 14 ft on each side and a slant height of 10 ft. Shingles cover only the slanted faces. One bundle covers 35 ft
- ²
- . How many bundles are needed?

Answer _____ bundles

- 4.
- Lampshade.**
- A lampshade is a square pyramid with no base. Its lateral area is 160 in
- ²
- and each base edge is 8 in. What is its slant height?

Answer _____ in

- 5.
- Play Tent.**
- A play tent is a square pyramid with a base 5 ft on each side and a slant height of 6 ft. It has no floor. Sasha says it needs 85 ft
- ²
- of fabric. Is she right? Explain.

TEST PRACTICE*choose the best answer*

6. A square pyramid has a base 4 cm on each side and a slant height of 7 cm. What is its lateral area?

A 14 cm²**B** 56 cm²**C** 72 cm²**D** 112 cm²

7. A pyramid has a surface area of 120 m
- ²
- and a lateral area of 84 m
- ²
- . What is its base area?

A 204 m²**B** 120 m²**C** 36 m²**D** 42 m²**CHECK YOURSELF**
☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____